

Exercise 5 Advanced SQL NULL functions

1. Syntax `SELECT employee_id, name,
IFNULL(salary, '0') As salary_with_default
FROM employees;`

employees_table

employee_id	name	salary_with_default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

2. Syntax `SELECT employee_id, name
IFNULL(department, 'not assigned') As department_name
FROM employees;`

employees_table

employees_id	name	department_name
1	Alice	HR
2	Bob	IT
3	Charlie	not assigned
4	Dana	Finance

3. Syntax `SELECT order_id, customer_id
FROM orders_table
WHERE customer_id IS NULL;`

orders_table

order_id	customer_id
103	NULL

102

103

4. Syntax `SELECT order-id, customer-id,
IFNULL (delivery-date, 'pending') As delivery-status
FROM orders-table;`

Orders-table

order-id	customer-id	delivery-status
101	201	2024-12-01
102	202	pending
103	NULL	2024-12-03

5. Syntax `SELECT student-id, name,
IFNULL (grade, '0') As graded final-grade
FROM students-table;`

students-table

student-id	name	Final-grade
1	Ethan	85
2	Maya	0
3	Oliver	90

6. Syntax `SELECT COUNT (grade) As not-graded-count
FROM students-table
WHERE grade ISNULL;`

students-table
not-graded-count
1

7. Syntax `SELECT product-id, name,
IFNULL (price, discount, '0') As final-price
FROM products-table;`

products-table

product-id	name	Final-price
501	keyboard	25
502	mouse	10
503	monitor	100

8. Syntax `SELECT COUNT(email) AS missing_email_count`
`FROM customers_table`
`WHERE email IS NULL;`

customers_table

missing_email_count
2

9. Syntax `SELECT customer_id, name,`
`IFNULL(email, 'No email') AS email_display`
`FROM customers_table;`

customers_table

customer_id	name	email_display
1	Linda	No email
2	Joseph	joseph@gmail.com
3	Nia	No email

10. Syntax `SELECT payment_id, status,`
`IFNULL(method, 'unknown') AS method_display`
`FROM payments_table;`

payments

payment_id	method_display	status
301	Credit	NULL
302	Paypal	Success
303	unknown	Failed

11. Syntax `SELECT item_id, item_name,`
`IFNULL(quantity, '0') AS quantity_checked`
`FROM inventory_table;`

Inventory_table

item_id	item_name	quantity_checked
1	Pen	0
2	Notebook	150
3	Eraser	0

12 Syntax SELECT emp-id

COALESCE(bonus, commission, 0) As first-available-reward
FROM employees-extra-table;

employees-extra-table

emp-id	first-available-reward
1	300
2	100
3	0

13 Syntax SELECT COUNT(room) As no-room-count

FROM classes

WHERE room ISNULL;

classes-table

no-room-count
2

14 Syntax SELECT student-id, date

IFNULL(status, 'Not marked') As attendance-status

FROM attendance-table;

attendance-table

student-id	date	attendance-status
1	2025-04-01	Not marked.
2	2025-04-01	Present
3	2025-04-01	Absent

15. Syntax: SELECT account-id,

IFNULL(account-type, 'Unknown') As type-display

IFNULL(balance, '0') As balance checked

FROM bank-accounts-table;

bank-accounts-table

account-id	type-display	balance-checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

16. Syntax `SELECT project-id, title
IFNULL (start-date, 'TBD') AS start_display
FROM projects-table;`

Projects-table

project-id	title	start_display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

17. Syntax `SELECT review-id, product-id,
IFNULL (comment, 'No comment') AS comment_display
IFNULL (rating, '0') AS rating_display
FROM reviews-table;`

reviews-table

review-id	product-id	comment_display	rating_display
1	501	Great product	4
2	502	No comment	0
3	503	Works fine	3

18. Syntax `SELECT supplier-id, name,
COALESCE (phone, alt-phone, 'No contact') AS contact_number
FROM suppliers-table;`

Suppliers-table

Supplier-id	name	contact_number
1	Global Goods	123456789
2	Best Supplies	987654321
3	ValueSource	No contact

19 Syntax SELECT user_id,
 IFNULL(theme, 'light') As theme_set,
 IFNULL(language, 'English') As language_set,
 IFNULL(timezone, 'UTC') As time_zone_set
 FROM user_settings_table;

User_Settings-table

user_id	theme_set	language_set	timezon_set
1	light	English	UTC
2	Dark	English	UTC+1
3	Light	English	UTC

20 Syntax SELECT record_id, machine_id,
 IFNULL(issue, 'Unknown Issue') As Issue_log
 IFNULL(technician, 'Not assigned') As technician_name
 FROM maintenance_table;

Maintenance

record_id	machine_id	issue_log	technician_log
1	M101	Overheating	Not assigned
2	M102	Unknown Issue	Not assigned
3	M103	Jammed	Alex